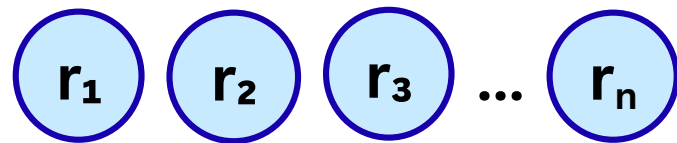


1 FluxTransformer model

Reaction tokens



FluxTransformer

Predicted fluxes $\hat{\mathbf{v}}$

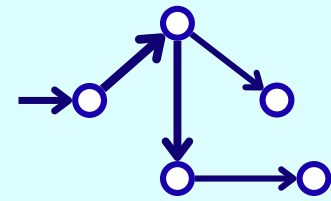
$\hat{v}_1$ $\hat{v}_2$ $\hat{v}_3$... $\hat{v}_n$

2 Generate FBA-simulated data

Sampled nutrient bounds

$b_1^{(j)}$ $b_2^{(j)}$... $b_n^{(j)}$

FBA/pFBA



Flux solution $\mathbf{v}^{(j)}$

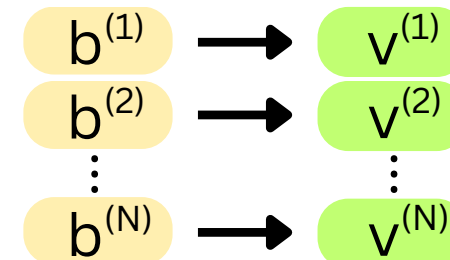
$v_1^{(j)}$ $v_2^{(j)}$... $v_n^{(j)}$

Training pair

$(\mathbf{b}^{(j)}, \mathbf{v}^{(j)})$

3 Pretrain on simulated data

Training examples



FluxTransformer

Prediction

$\hat{\mathbf{v}}$

FBA target

$\mathbf{v}$

Update weights

Loss

4 Transfer to experimental tasks

Growth / Omics data

MLP

$b_1^{(j)}$ $b_2^{(j)}$... $b_n^{(j)}$

FluxTransformer



Predicted full flux distribution $\hat{\mathbf{v}}$

$\hat{v}_1$ $\hat{v}_2$ $\hat{v}_3$... $\hat{v}_n$